Sequences and Basic Series Convergence Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Limits of Sequences & Monotonicity**

Find the interval of convergence of $\sum_{n=0}^{\infty} x^n$.

Theorem (Interval of Convergence for Geometric Power Series): The series $\sum_{n=0}^{\infty} x^n$ is a geometric series with common ratio $r = x$. Step 1 · Apply Geometric Convergence Criterion: An infinite geometric series converges if and only if $|r| < 1$. Therefore: $|x| < 1 \Rightarrow -1 < x < 1$. Radius of convergence is $R = 1$, centered at $c = 0$. Step 2 · Test the boundary endpoints: At $x = 1$: $\sum 1^n = 1 + 1 + 1 + \ldots$ diverges by the nth-term test ($\lim_{n\to\infty} 1 = 1e0$). At $x = -1$: $\sum (-1)^n = 1 - 1 + 1 - 1 + \ldots$ diverges by oscillation ($\lim_{n\to\infty} (-1)^n$ DNE). Conclusion: Neither endpoint is included. The interval of convergence is strictly: $\mathbf{(-1, 1)}$.

2 **[Foundational] • Target: Geometric Series & n-th Term Divergence Test**

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

The definite integral $\int_{-2}^{2} f(x)\,dx$ represents the net signed area between the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$. The graph of $f(x)$ for $-2 \le x \le 2$ forms a semicircle above the x-axis with radius 2. The area of this semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$. However, the provided figure shows a region that is a combination of triangles and a rectangle. From $x = -2$ to $x = 0$, the area is a triangle with base 2 and height 2, so the area is $\frac{1}{2} \times 2 \times 2 = 2$. From $x = 0$ to $x = 2$, the area is a triangle with base 2 and height 2, so the area is $\frac{1}{2} \times 2 \times 2 = 2$. The total net signed area is $2 + 2 = 4$.

3 **[Foundational] • Target: Integral Test & p-Series**

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

Method · Taylor/Maclaurin Polynomial Approximation: Step 1 · Define the two-term (degree 1) Maclaurin polynomial for e^x: $P_1(x) = 1 + x$. Step 2 · Evaluate at $x = 0.1$: $P_1(0.1) = 1 + 0.1 = \mathbf{1.1}$. Step 3 · Error analysis via Lagrange Remainder: By Taylor's Theorem:
$|R_1(0.1)| = \left|\frac{f''(c)}{2!}(0.1)^2\right| = \frac{e^c}{2}(0.01)$ for some $c \in (0, 0.1)$. Since $e^c < e^{0.1} < 1.2$,
$|R_1(0.1)| < \frac{1.2}{2}(0.01) = 0.006$. Exact value: $e^{0.1} \approx 1.10517$. The two-term approximation $\mathbf{1.1}$ captures the true value within 0.47%.

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4 [Foundational] • Target: Direct Comparison & Limit Comparison Tests

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$.

Theorem (Ratio Test for Radius of Convergence): Step 1 · Set up ratio limit for $a_n = \frac{x^n}{2^n}$: $L = \lim_{n\to\infty} \left|\frac{x^{n+1}}{2^{n+1}} \cdot \frac{2^n}{x^n}\right| = \lim_{n\to\infty} \left|\frac{x \cdot 2^n}{2 \cdot 2^n}\right| = \frac{|x|}{2}$. Step 2 · Apply convergence condition $L < 1$: $\frac{|x|}{2} < 1 \Rightarrow |x| < 2$. Conclusion: The series converges for all x strictly within distance 2 of the center $c = 0$. The radius of convergence is: $\mathbf{R = 2}$.

5 [Foundational] • Target: Limits of Sequences & Monotonicity

Consider the function represented by the graph of $f(x)$ below. What is the value of the definite integral $\int_0^4 f(x)\,dx$?

The definite integral $\int_0^4 f(x)\,dx$ represents the area under the curve $f(x)$ from $x = 0$ to $x = 4$. The graph of $f(x)$ from $x = 0$ to $x = 4$ forms a triangle with a base of 4 units (from 0 to 4 on the x-axis) and a height of 6 units (the value of $f(x)$ at $x = 0$). The area of a triangle is given by $\frac{1}{2} \times \text{base} \times \text{height}$. Therefore, the area is $\frac{1}{2} \times 4 \times 6 = 12$.

6 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

Consider the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. For what value(s) of p does this series converge?

The given series is the p-series, $\sum_{n=1}^{\infty} \frac{1}{n^p}$. This series is a fundamental example in the study of convergence. The p-series test states that the p-series converges if and only if $p > 1$. Therefore, the series converges for $p > 1$.

7 [Foundational] • Target: Integral Test & p-Series

Find the arc length of $x = t,\ y = t$ for $t \in [0, 1]$.

Theorem (Parametric Arc Length Formula): If a curve is given parametrically by $x(t)$ and $y(t)$ with continuous derivatives on $[a, b]$, its arc length L is: $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$. Step 1 · Compute derivatives: $\frac{dx}{dt} = \frac{d}{dt}[t] = 1, \quad \frac{dy}{dt} = \frac{d}{dt}[t] = 1$. Step 2 · Evaluate the integrand (differential speed element): $\sqrt{(1)^2 + (1)^2} = \sqrt{1 + 1} = \sqrt{2}$. Step 3 · Integrate over $t \in [0, 1]$: $L = \int_0^1 \sqrt{2}\,dt = [\sqrt{2}\,t]_0^1 = \sqrt{2}(1) - \sqrt{2}(0) = \sqrt{\mathbf{2}}$. Geometric check: The curve represents the straight line segment connecting $(0, 0)$ to $(1, 1)$. By the Pythagorean distance formula, $d = \sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Method · Telescoping Series Evaluation via Partial Fractions: Step 1 · Decompose general term: $\frac{1}{N(N+1)} = \frac{1}{N} - \frac{1}{N+1}$. Step 2 · Write out the Nth partial sum S_N: $S_N = \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ \dots \ + \left(\frac{1}{N} - \frac{1}{N+1}\right)$. Step 3 · Cancel intermediate terms: Every internal term cancels symmetrically, leaving only the first and last components: $S_N = 1 - \frac{1}{N+1}$. Step 4 · Take the limit as $N \to \infty$: $S = \lim_{N\to\infty} S_N = \lim_{N\to\infty} \left(1 - \frac{1}{N+1}\right) = 1 - 0 = \mathbf{1}$.

9 [Intermediate] • Target: Limits of Sequences & Monotonicity

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge?

Theorem (Alternating Series Test / Leibniz's Theorem): An alternating series of the form $\sum_{n=1}^{\infty} (-1)^n b_n$ or $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ converges if all three conditions are satisfied: 1. $b_n > 0$ for all n; 2. $b_{n+1} \le b_n$ for all $n \ge N$ (sequence of magnitudes is monotonically decreasing); 3. $\lim_{n\to\infty} b_n = 0$. Verification for $b_n = \frac{1}{n}$: 1. $b_n = \frac{1}{n} > 0$ for all integers $n \ge 1$. 2. $b_{n+1} = \frac{1}{n+1} < \frac{1}{n} = b_n$ holds for all $n \ge 1$. 3. $\lim_{n\to\infty} \frac{1}{n} = 0$. Conclusion: All three hypotheses are met, so the series converges (conditionally, with exact sum $-\ln 2$).

10 [Intermediate] • Target: Geometric Series & n-th Term Divergence Test

Find the speed of a particle with $x = t^2,\ y = t^2$ at $t = 1$.

Theorem (Particle Kinematics & Speed Formula): For a particle whose position vector is $\vec{r}(t) = \langle x(t), y(t) \rangle$, the velocity vector is $\vec{v}(t) = \langle x'(t), y'(t) \rangle$, and its instantaneous speed is the Euclidean magnitude of velocity: $\text{Speed} = ||\vec{v}(t)|| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Step 1 · Determine velocity components: $x'(t) = \frac{d}{dt}[t^2] = 2t, \quad y'(t) = \frac{d}{dt}[t^2] = 2t$. Step 2 · Evaluate components at $t = 1$: $x'(1) = 2(1) = 2, \quad y'(1) = 2(1) = 2$. Step 3 · Compute speed: $\text{Speed}(1) = \sqrt{(2)^2 + (2)^2} = \sqrt{4+4} = \sqrt{8} = \mathbf{2\sqrt{2}}$.

11 [Intermediate] • Target: Integral Test & p-Series

Evaluate $\int \frac{1}{x^2 - 1}\, dx$.

Method · Partial Fraction Decomposition (Distinct Linear Factors): Step 1 · Factor the denominator: $\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$. Step 2 · Solve for coefficients via Heaviside Cover-Up: $A = \frac{1}{x+1}\Big|_{x=1} = \frac{1}{1+1} = \frac{1}{2}, \quad B = \frac{1}{x-1}\Big|_{x=-1} = \frac{1}{-1-1} = -\frac{1}{2}$. Step 3 · Integrate term by term: $\int \frac{1}{x^2-1}\, dx = \frac{1}{2}\int \frac{1}{x-1}\, dx - \frac{1}{2}\int \frac{1}{x+1}\, dx = \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$. Step 4 · Combine logarithms: $\mathbf{\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C}$.

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12 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge?

Theorem (Divergence of the Harmonic Series): The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the fundamental Harmonic Series. Proof via p-Series Test: A p-series $\sum \frac{1}{n^p}$ converges if and only if $p > 1$. Here, $p = 1 \leq 1$, so the series diverges. Alternative Proof via Integral Test: The function $f(x) = \frac{1}{x}$ is positive, continuous, and strictly decreasing on $[1, \infty)$: $\int_1^{\infty} \frac{1}{x}\,dx = \lim_{b\to\infty} [\ln x]_1^b = \lim_{b\to\infty} (\ln b - 0) = \infty$. Because the improper integral diverges, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. Conclusion: Diverges.

13 [Intermediate] • Target: Limits of Sequences & Monotonicity

Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?

Theorem (p-Series Convergence Test): The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$, and diverges if $p \leq 1$. Step 1 · Identify the series parameter p: The general term is $a_n = \frac{1}{n^2}$, which is a p-series with exponent $p = 2$. Step 2 · Test against convergence condition: Because $p = 2 > 1$, the series converges by the p-Series Test. Analytical Note: By Euler's classical Basel solution, this series converges to the exact sum: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.64493$. Conclusion: Converges by the **p**−Series Test **(p = 2 > 1)**.

14 [Intermediate] • Target: Geometric Series & n-th Term Divergence Test

Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge?

Theorem (nth-Term Divergence Test): If $\lim_{n\to\infty} a_n e 0$ or does not exist, then the infinite series $\sum_{n=1}^{\infty} a_n$ diverges. Step 1 · Evaluate the limit of the general term as $n \to \infty$:

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = \frac{1}{1+0} = 1$. Step 2 · Apply the test: Since $\lim_{n\to\infty} a_n = 1 e 0$, the sequence of terms does not approach zero. Therefore, the series cannot converge. Conclusion:

Diverges by the **n**th – Term Divergence Test.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$.

Theorem (Parametric First Derivative): For parametric equations $x = x(t)$ and $y = y(t)$:

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{dy/dt}{dx/dt}$, provided $\frac{dx}{dt} e 0$. Step 1 · Differentiate with respect to t:

$\frac{dx}{dt} = \frac{d}{dt}[t^2] = 2t$, $\frac{dy}{dt} = \frac{d}{dt}[t^3] = 3t^2$. Step 2 · Form the quotient: $\frac{dy}{dx} = \frac{3t^2}{2t} = \mathbf{\frac{3}{2}t}$ $(te0)$. Verification via Cartesian Equation: $x = t^2 \Rightarrow t = x^{1/2} \Rightarrow y = (x^{1/2})^3 = x^{3/2}$. Then $\frac{dy}{dx} = \frac{3}{2}x^{1/2} = \frac{3}{2}t$, in exact agreement.

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16 **[Challenging]** • **Target: Direct Comparison & Limit Comparison Tests** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

Evaluate $\int_0^\infty e^{-x}\,dx$.

Theorem (Improper Exponential Integral): Formulate the integral as the limit of proper definite integrals: $\int_0^\infty e^{-x}\,dx = \lim_{b} \int_0^b e^{-x}\,dx$. Step 1 · Antidifferentiate: $\int_0^b e^{-x}\,dx = [-e^{-x}]_0^b = (-e^{-b}) - (-e^0) = 1 - e^{-b} = 1 - \frac{1}{e^b}$. Step 2 · Evaluate the limit: Since $\lim_{b\to\infty} e^b = \infty$, we have $\lim_{b\to\infty} \frac{1}{e^b} = 0$: $\lim_{b\to\infty} \left(1 - \frac{1}{e^b}\right) = 1 - 0 = \mathbf{1}$. Conclusion: The improper integral converges to $\mathbf{1}$.

17 **[Challenging]** • **Target: Limits of Sequences & Monotonicity** • **Cross-Lesson: 3.01 Parametric Equations**

Evaluate $\int x\,e^{x^2}\,dx$.

Method · Integration by Substitution (u-Substitution): Step 1 · Identify inner function and differential: Notice that the derivative of x^2 is $2x$, and the integrand contains the factor $x\,dx$. Let: $u = x^2 \Rightarrow du = 2x\,dx \Rightarrow x\,dx = \frac{1}{2}\,du$. Step 2 · Substitute into the integral: $\int x e^{x^2}\,dx = \int e^{x^2}(x\,dx) = \int e^u \left(\frac{1}{2}\,du\right) = \frac{1}{2}\int e^u\,du$. Step 3 · Integrate and substitute back $u = x^2$: $\frac{1}{2}e^u + C = \mathbf{\frac{1}{2}e^{x^2} + C}$.

18 **[Challenging]** • **Target: Geometric Series & n-th Term Divergence Test** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Evaluate $\sum_{n=0}^{\infty} (\frac{1}{2})^n$.

Theorem (Geometric Series Sum Formula): For an infinite geometric series $\sum_{n=0}^{\infty} ar^n$, the series converges if and only if $|r| < 1$, and its sum is given by: $S = \frac{a}{1-r}$. Step 1 · Identify parameters: Initial term ($n = 0$): $a = \left(\frac{1}{2}\right)^0 = 1$. Common ratio: $r = \frac{1}{2}$. Step 2 · Check convergence criterion: $|r| = |1/2| = 0.5 < 1$, so the series converges. Step 3 · Compute the sum: $S = \frac{1}{1 - 1/2} = \frac{1}{1/2} = \mathbf{2}$.

19 **[Challenging]** • **Target: Integral Test & p-Series** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Evaluate $\int x\,e^x\,dx$.

Theorem (Integration by Parts Formula): For continuously differentiable functions $u(x)$ and $v(x)$: $\int u\,dv = uv - \int v\,du$. Step 1 · Choose u and dv via LIATE protocol: Algebraic (x) precedes Exponential (e^x), so set: $u = x \Rightarrow du = dx, \quad dv = e^x\,dx \Rightarrow v = \int e^x\,dx = e^x$. Step 2 · Apply the formula: $\int x e^x\,dx = xe^x - \int e^x\,dx$. Step 3 · Integrate the remaining term: $xe^x - e^x + C = \mathbf{(x-1)e^x + C}$.

20 **[Challenging]** • **Target: Direct Comparison & Limit Comparison Tests** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

For $x = e^t,\ y = e^{2t}$, find $\frac{dy}{dx}$.

Method 1 · Parametric Derivative: Step 1 · Differentiate both components with respect to t: $\frac{dx}{dt} = \frac{d}{dt}[e^t] = e^t, \quad \frac{dy}{dt} = \frac{d}{dt}[e^{2t}] = 2e^{2t}$. Step 2 · Compute derivative ratio: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2e^{2t}}{e^t} = 2e^{2t-t} = \mathbf{2e^t}$. Method 2 · Cartesian Elimination: Notice $y = e^{2t} = (e^t)^2 = x^2$. Differentiating directly with respect to x: $\frac{dy}{dx} = 2x = \mathbf{2e^t}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits of Sequences & Monotonicity

Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2 ✓

b. 4

c. ∞ (diverges)

d. 0

Divide numerator and denominator by n^2: $\lim_{n\to\infty} \frac{4 - 3/n}{2 + 5/n^2} = \frac{4}{2} = 2$.

22 [Foundational] • Target: Limits of Sequences & Monotonicity

Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. **Every bounded monotonic sequence converges** ✓

b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

The Monotone Convergence Theorem states that if a sequence is bounded and monotonic (either non-decreasing or non-increasing), it must converge.

23 [Foundational] • Target: Limits of Sequences & Monotonicity

What is $\lim_{n\to\infty} (1 + \frac{2}{n})^n$?

a. e^2 ✓

b. e

c. 1

d. ∞

By the standard exponential limit $\lim_{n\to\infty} (1 + \frac{x}{n})^n = e^x$, with $x = 2$, the limit is e^2.

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24 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.

a. 2 ✓

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

First term $a = 3(\frac{2}{5}) = \frac{6}{5}$, common ratio $r = \frac{2}{5}$. Sum $= \frac{a}{1-r} = \frac{6/5}{1-2/5} = \frac{6/5}{3/5} = 2$.

25 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. **The series diverges** ✓

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

The n-th Term Divergence Test states that if $\lim_{n\to\infty} a_n \neq 0$, the series must diverge.

26 [Foundational] • Target: Geometric Series & n-th Term Divergence Test

If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. **The test is inconclusive; the series may converge or diverge** ✓

b. The series converges

c. The series diverges

d. The sum is zero

Having terms approach zero is a necessary condition for convergence, but not sufficient (e.g., the harmonic series $\sum \frac{1}{n}$ has terms approaching 0 but diverges).

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27 [Foundational] • Target: Integral Test & p-Series

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

By the p-series test, $\sum \frac{1}{n^p}$ converges if and only if $p > 1$. For $p \leq 1$, it diverges.

28 [Foundational] • Target: Integral Test & p-Series

Which series converges by the Integral Test?

a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ ✓

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = [-\frac{1}{\ln x}]_2^{\infty} = \frac{1}{\ln 2} < \infty$. Since the improper integral converges, the series converges.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integral Test & p-Series

What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ **must be continuous, positive, and decreasing** ✓

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

The hypotheses of the Integral Test require $f(x)$ to be continuous, positive, and decreasing on $[1, \infty)$ where $f(n) = a_n$.

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30 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$ ✓

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

For all $n \geq 1$: $\frac{1}{n^3+5n} < \frac{1}{n^3}$. Since $\sum \frac{1}{n^3}$ is a convergent p-series ($p = 3 > 1$), the given series converges by Direct Comparison.

31 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ (convergent) ✓

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

The dominant terms are $\frac{3n}{2n^3} = \frac{3}{2n^2}$. Comparing with $b_n = \frac{1}{n^2}$ yields a positive finite limit $\frac{3}{2}$, confirming convergence.

32 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge ✓

b. $\sum a_n$ converges to L

c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

A finite positive limit L ensures that both series share the exact same convergence or divergence behavior.

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33 **[Intermediate]** • **Target: Limits of Sequences & Monotonicity**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+6}$ converges.

a. Diverges by the nth – term test

b. Converges (compare to $\sum 1/n^2$) ✓

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

Since 01), the given series converges by the comparison test.

34 **[Intermediate]** • **Target: Geometric Series & n-th Term Divergence Test**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+5}$ converges.

a. Inconclusive

b. Diverges by the nth – term test

c. Diverges (compare to $\sum 1/n$)

d. Converges (compare to $\sum 1/n^2$) ✓

Since 01), the given series converges by the comparison test.

35 **[Intermediate]** • **Target: Integral Test & p-Series**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges.

a. Diverges (compare to $\sum 1/n$)

b. Inconclusive

c. Diverges by the nth – term test

d. Converges (compare to $\sum 1/n^2$) ✓

Since 01), the given series converges by the comparison test.

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36 [Intermediate] • Target: Direct Comparison & Limit Comparison Tests

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+2}$ converges.

a. Converges (compare to $\sum 1/n^2$) ✓

b. Diverges by the nth – term test

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

Since 01), the given series converges by the comparison test.

37 [Intermediate] • Target: Limits of Sequences & Monotonicity

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converges.

a. Diverges (compare to $\sum 1/n$)

b. Diverges by the nth – term test

c. Converges (compare to $\sum 1/n^2$) ✓

d. Inconclusive

Since 01), the given series converges by the comparison test.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Geometric Series & n-th Term Divergence Test • Cross-Lesson: 3.02 Vector-Valued Functions

Consider the infinite series $\sum_{n=1}^{\infty} a_n$. If the terms of the series are given by $a_n = \frac{3n+1}{2n+5}$, which of the following statements about the convergence of the series is true based on the limit of the terms?

a. The series converges because $\lim_{n\to\infty} a_n = 0$.

b. **The series diverges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.** ✓

c. The series converges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

d. The series diverges because $\lim_{n\to\infty} a_n$ does not exist.

To determine the convergence or divergence of an infinite series $\sum_{n=1}^{\infty} a_n$, we can first examine the limit of its terms, $\lim_{n\to\infty} a_n$. If this limit is not equal to zero, then the series diverges by the Test for Divergence. For the given series, the terms are $a_n = \frac{3n+1}{2n+5}$. We evaluate the limit as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n+1}{2n+5}$ To evaluate this limit, we can divide both the numerator and the denominator by the highest power of n in the denominator, which is n: $\lim_{n\to\infty} \frac{\frac{3n}{n}+\frac{1}{n}}{\frac{2n}{n}+\frac{5}{n}} = \lim_{n\to\infty} \frac{3+\frac{1}{n}}{2+\frac{5}{n}}$ As $n \to \infty$, $\frac{1}{n} \to 0$ and $\frac{5}{n} \to 0$. Therefore, the limit becomes: $\frac{3+0}{2+0} = \frac{3}{2}$ Since $\lim_{n\to\infty} a_n = \frac{3}{2}$, and $\frac{3}{2} eq 0$, the series diverges by the Test for Divergence. Therefore, choice B is correct.

39 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{5^n}$. The series converges if and only if the parameter p satisfies which of the following conditions? Use the Ratio Test to determine the interval of convergence.

a. $p > 0$

b. $p < 1$

c. $p > -1$

d. **For all real values of p** ✓

We apply the Ratio Test. Let $a_n = \frac{n^p}{5^n}$. Then $a_{n+1} = \frac{(n+1)^p}{5^{n+1}}$. We compute the limit of the ratio of consecutive terms: $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)^p}{5^{n+1}} \cdot \frac{5^n}{n^p}\right| L = \lim_{n\to\infty} \left|\frac{1}{5}\left(\frac{n+1}{n}\right)^p\right| = \lim_{n\to\infty} \left|\frac{1}{5}\left(1+\frac{1}{n}\right)^p\right|$ As $n \to \infty$, $\frac{1}{n} \to 0$, so $\left(1+\frac{1}{n}\right)^p \to 1^p = 1$ for any real p. Thus, $L = \frac{1}{5} \cdot 1 = \frac{1}{5}$. According to the Ratio Test, the series converges if $L < 1$. Since $L = \frac{1}{5} < 1$ for all real values of p, the series converges for all real values of p.

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40 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 4.02 Area in Polar Coordinates

Consider the infinite sequence defined by $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$. To determine if this sequence converges, we examine its limit as $n \to \infty$. Which of the following statements is true about the limit of a_n?

a. The limit is 0.

b. The limit is 2.

c. **The limit is 3.** ✓

d. The sequence diverges.

To find the limit of the sequence $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$ as $n \to \infty$, we can divide both the numerator and the denominator by the highest power of n in the denominator, which is n^2. x As $n \to \infty$, the terms $\frac{2}{n}$, $\frac{1}{n^2}$, and $\frac{5}{n^2}$ all approach 0. $= \frac{3 + 0 - 0}{1 + 0} = \frac{3}{1} = 3$ Therefore, the sequence converges to 3.

41 [Challenging] • Target: Limits of Sequences & Monotonicity • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Geometric series $\sum ar^n$ converges when:

a. $r > 1$

b. $|r| < 1$ ✓

c. $r < 0$

d. $r = 1$

Applying Unit 10 · Infinite Sequences & Series — Convergence Tests principles, Convergence requires $|r| < 1$. Therefore, the verified correct option is $|r| < 1$.

42 [Challenging] • Target: Geometric Series & n-th Term Divergence Test • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For $\sum_{n=1}^{\infty} \frac{n^{13}}{2^n}$, apply the ratio test and determine convergence.

a. Converges ($L = 1/2 < 1$) ✓

b. Inconclusive ($L = 1$)

c. Diverges by comparison

d. Diverges ($L = 2 > 1$)

$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^{13} \cdot \frac{1}{2} \to \frac{1}{2} < 1$. Ratio test: converges regardless of the fixed power 13.

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43 [Challenging] • Target: Integral Test & p-Series • Cross-Lesson: 6.02 Taylor and Maclaurin Series

For $\sum_{n=1}^{\infty} \frac{n^{12}}{2^n}$, apply the ratio test and determine convergence.

a. Diverges by comparison

b. Diverges ($L = 2 > 1$)

c. Converges ($L = 1/2 < 1$) ✓

d. Inconclusive ($L = 1$)

$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^{12} \cdot \frac{1}{2} \to \frac{1}{2} < 1$. Ratio test: converges regardless of the fixed power 12.

44 [Challenging] • Target: Direct Comparison & Limit Comparison Tests • Cross-Lesson: 1.01 Integration by Parts

For $\sum_{n=1}^{\infty} \frac{n^{10}}{2^n}$, apply the ratio test and determine convergence.

a. Diverges by comparison

b. Inconclusive ($L = 1$)

c. Diverges ($L = 2 > 1$)

d. Converges ($L = 1/2 < 1$) ✓

$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^{10} \cdot \frac{1}{2} \to \frac{1}{2} < 1$. Ratio test: converges regardless of the fixed power 10.

45 [Challenging] • Target: Limits of Sequences & Monotonicity • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $\sum_{n=1}^{\infty} \frac{n^{11}}{2^n}$, apply the ratio test and determine convergence.

a. Diverges ($L = 2 > 1$)

b. Diverges by comparison

c. Inconclusive ($L = 1$)

d. Converges ($L = 1/2 < 1$) ✓

$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^{11} \cdot \frac{1}{2} \to \frac{1}{2} < 1$. Ratio test: converges regardless of the fixed power 11.

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